CS 4412: Data Mining - M3 Complete Implementation Report

Course: CS 4412: Data Mining

Student Name:

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1. Implementation Status

The analysis pipeline is functionally complete and runs end-to-end successfully. The following stages have been implemented:

- **Data Ingestion & Preprocessing:** The music_data.csv dataset (57,976 original rows) was successfully loaded. Missing values were imputed using the median, and a log-transformation was applied to the acoustic_electronic_ratio (creating acoustic_electronic_ratio_log) to handle extreme outliers. The data was then standardized using StandardScaler.
- **Data Cleaning:** Exact duplicate rows were identified and removed, reducing the dataset to 50,000 unique observations, which prevented centroid bias during clustering.
- **Mining Techniques Applied:**
  - **Clustering:** K-Means clustering was implemented and evaluated across a range of k values ($k = 2$ to $k = 8$).
  - **Dimensionality Reduction:** Principal Component Analysis (PCA) was implemented to visualize the high-dimensional clusters in a 2D space.
  - **Classification (for Discovery):** Decision Trees were implemented using the cluster assignments as class labels to extract interpretable rules about what distinguishes the clusters.
  - **Anomaly Detection:** Both Local Outlier Factor (LOF) and DBSCAN algorithms were evaluated to identify unusual tracks within the dataset.

2. Analytical Choices

To ensure meaningful analytical expansion beyond M2, several techniques and parameters were explored:

- **Clustering Evaluation:** Instead of arbitrarily choosing a k value, the Elbow Method, Silhouette Score, and Davies-Bouldin Index were computed for $k \in [2,8]$. Based on the metrics and the goal to discover nuanced segments, $k = 4$ was selected as the optimal number of clusters.
- **Classification for Interpretability:** A Decision Tree Classifier was used not for predictive accuracy, but for pattern discovery. By fitting a shallow tree (depth=4) to the K-Means cluster labels, the model acted as an interpretability tool. This revealed exactly which feature thresholds naturally separate the music segments.
- **Dimensionality Reduction:** PCA was chosen primarily for visualization purposes. Reducing the 8-feature space down to two principal components (which captured

36.24% of the total variance) allowed for visual validation of the cluster separations and outlier detection.

- **Anomaly Detection Comparison:** Both distance-based and density-based methods were explored. DBSCAN (eps=0.5, min_samples=10) was run but struggled with the high-dimensional density of the standardized data, flagging all 50,000 points as noise. This highlighted the advantage of LOF, which identifies points in low-density regions *relative* to their local neighbors. LOF proved much more robust for these varying cluster densities, successfully isolating a meaningful 5% anomaly rate.

3. Results and Pattern Discovery

The end-to-end pipeline yielded several substantial findings:

- **Cluster Characteristics & Feature Importance:** The Decision Tree revealed that just two features almost entirely dictate the cluster splits: acoustic_electronic_ratio_log (57.4% importance) and instrumentalness (41.1%). This indicates that the balance of acoustic versus electronic elements, alongside the presence or absence of vocals, define the core mathematical differences between music segments in this dataset.
- **Cluster Sizes:** The deduplicated data segmented into four distinct groups. Cluster 0 is the largest (mainstream/balanced with 24,715 tracks), while Cluster 3 is a highly specific niche group (size: 205).
- **Anomaly Distribution:** Applying LOF revealed that anomalies are not distributed evenly across the music types.
  - Cluster 0: 2.9% anomalies (717 / 24,715)
  - Cluster 1: 9.3% anomalies (810 / 8,716)
  - Cluster 2: 5.7% anomalies (936 / 16,364)
  - Cluster 3: 18.0% anomalies (37 / 205)
  - *Interpretation:* The tiny niche group (Cluster 3) contains a vastly disproportionate amount of unusual tracks (18%) compared to the mainstream groups, suggesting that edge-case music production is highly concentrated in this specific auditory profile.

4. Discovery Question Progress

- *What types/segments of music exist?* Addressed via K-Means ($k = 4$). We successfully identified 4 core auditory profiles in the dataset.
- *What features distinguish these groups?* Addressed via Decision Trees. The analysis confirmed that the acoustic/electronic log ratio and instrumentalness are the absolute dominant defining factors, overriding features like tempo, valence, or loudness.
- *What's unusual or anomalous?* Addressed via LOF. We discovered that anomalies are highly dependent on the sub-genre, with the smallest niche cluster exhibiting the highest rate of irregular feature combinations.

5. M4 Plan

The technical heavy lifting and functional code implementations are complete. The plan for M4 includes:

- **Final Polish:** Reformatting the Jupyter Notebook to be fully presentation-ready, ensuring all charts have clear legends, titles, and domain-specific annotations.
- **Critical Assessment:** Evaluating the limitations of the chosen models. For example, addressing the relatively low variance explained by the first two PCA components (36.24%) and discussing how that impacts the 2D visual representation of the clusters.
- **Domain Translation:** Translating the statistical anomaly findings into deeper domain language (e.g., investigating *why* Cluster 3 tracks are flagged heavily by LOF—are they data collection errors, or highly avant-garde tracks?).